

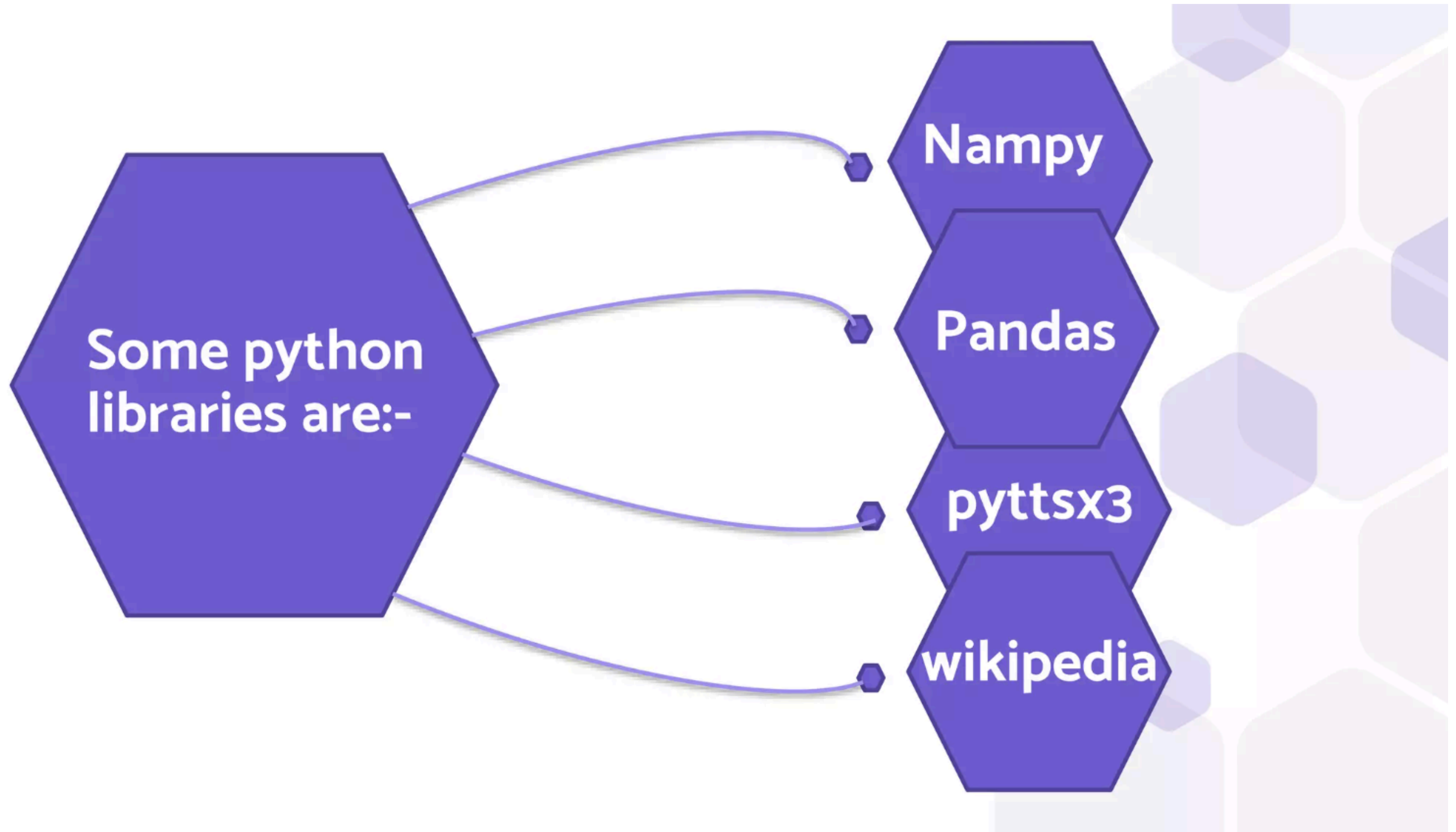
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Python Assignment-3

Python Library and Modules



Python Library

A library is a collection of modules. It is a reusable chunk of code that you may want to include in your programs/projects.



Numpy

Numpy is considered as one of the most popular machine learning library in Python. Array interface is the best and the most important feature of Numpy. This interface can be utilized for expressing images, sound waves, and other binary raw streams as an array of real numbers in N-dimensional.

Example 1-

```
import numpy as np  
arr = np.array([1, 2, 3, 4, 5])  
print(arr)  
print(type(arr))
```

Output- [1 2 3 4 5]
 <class 'numpy.ndarray'>

Example 2-

```
import numpy as np  
arr = np.array([[1, 2, 3], [4, 5, 6]])  
print(arr)
```

Output-
[[1 2 3]
 [4 5 6]]

Pandas

Pandas is a Python library used for working with data sets. It has functions for analyzing, cleaning, exploring, and manipulating data. It allows us to analyze big data and make conclusions based on statistical theories. It can clean messy data sets, and make them readable and relevant.

Example1

```
import pandas as pd
data = {'Name':[' Janny ', 'Amit','Suji']
        'Age':[28,34,29,42]}
df = pd.DataFrame(data)
print df
```

Its **output** is as follows –

	Age	Name
0	28	Janny
1	34	Amit
2	29	Suji

Pytttsx3

pytttsx3 is a text-to-speech conversion library in Python. Unlike alternative libraries, it works offline.

```
import pytttsx3  
engine = pytttsx3.init()  
engine.say("I will speak this text")  
engine.runAndWait()
```


Wikipedia

Wikipedia is a Python library that makes it easy to access and parse data from Wikipedia. Search Wikipedia, get article summaries, get data like links and images from a page, and more.

Example1

```
import wikipedia
```

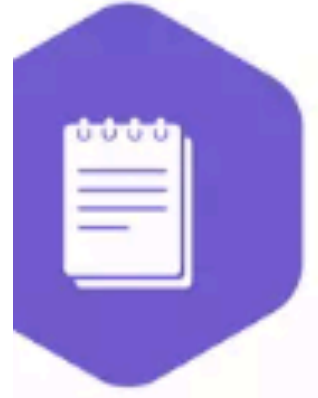
```
# finding result for the search
```

```
# sentences = 2 refers to numbers of line
```

```
result = wikipedia.summary("India", sentences = 2)
```

```
# printing the result
```

```
print(result)
```

Tkinter

Tkinter is a graphical user interface (GUI) module for Python, you can make desktop apps with Python and can develop **GUI** applications like **calculator**, **login system**, **text editor**, etc.



Tkinter Widgets

Tkinter provides various controls, such as buttons, labels and text boxes used in a GUI application. These controls are commonly called widgets.



Important method of Tkinter

1. `Tk(screenName=None, baseName=None, className='Tk', useTk=1)` :-

This method is mainly used **to create the main window**.

2. `The mainloop() Function`:-

This method is used to start the application. The `mainloop()` function is an **infinite loop** which is used to run the application, it will wait for **an event to occur** and **process the event** as long as the window is not closed.

Mini Project: Covid-19 Data Analysis

Github Link: <https://github.com/anshh2508/pythonassignment3>

Output:

```
=== COVID-19 Dataset ===
  Country  Total Cases  Total Deaths  Total Recovered
0    USA      34000000      609000      28000000
1   India      31000000      410000      30000000
2  Brazil      19000000      530000      17000000
3  Russia       6000000      150000       5800000
4     UK       5000000      130000       4800000

=== Dataset with Rates ===
  Country  Total Cases  ...  Death Rate (%)  Recovery Rate (%)
0    USA      34000000  ...      1.791176      82.352941
1   India      31000000  ...      1.322581      96.774194
2  Brazil      19000000  ...      2.789474      89.473684
3  Russia       6000000  ...      2.500000      96.666667
4     UK       5000000  ...      2.600000      96.000000

[5 rows x 6 columns]

=== Summary ===
USA has the highest number of total cases.
India has the highest recovery rate among the sample countries.
Death and recovery rates vary across different countries.
```


THANK YOU!